

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN

COURSE CODE: CSA1105

COURSE OUTCOME COVERED

***CO1:** Apply object-oriented concepts and software development life cycle models to design structured and modular software systems. (BL2, BL3)*

Assessment Tool Weightage: 30%

Dr Ramamoorthy

QUESTIONS: 6

Total Marks:30

Annexure A

SHORT ANSWER TEST

Code of Conduct:

*I __RASHMI NAURENE__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

rashmi

Annexure A

Short Answer Test

ANSWERS

1. Suppose a software system is developed for a **Smart Hotel Management System** where guests can book rooms, cancel reservations, and make online payments.

(a) Explain the following object-oriented concepts:

- Class- A class is a blueprint for creating objects. It contains data members (attributes) and methods (functions).
- Object- An object is an instance of a class. Each object has its own data and can use the methods defined in the class to perform functions.
- Inheritance- Inheritance is one of the properties of OOAD where a subclass inherits the properties and methods of a superclass. It promotes code reusability.
- Encapsulation- Encapsulation combines both the object and method such that unauthorized users cannot access the information without proper authorization
- Dynamic Binding- Dynamic binding decides which method to execute during runtime. It supports method overriding and polymorphism.

(b) Describe the following object relationships:

- Association- Association is a relationship where two objects work together. Both objects can exist independently.
- Dependency- Dependency means one class temporarily uses another class. Changes in one class may affect the dependent class.
- Composition- Composition is a strong relationship where one object owns another. If the parent object is destroyed, the child object is also destroyed.

(c) Explain how these concepts improve software maintainability and code reusability in a hotel management system.

ANSWER: Object-oriented concepts improve maintainability by making code modular and organized. They also increase reusability through inheritance and encapsulation.

(5 Marks)

2. Consider developing a **Smart Education Management System**.

(a) Explain the following object-oriented design principles:

- Abstraction- Abstraction hides unnecessary details and shows only essential features. It simplifies the system for users.
- Encapsulation- Encapsulation protects data by combining it with related methods. It prevents unauthorized access.
- Information Hiding- Information hiding hides internal implementation from other classes. Only required details are made accessible.

(b) Describe the following SOLID principles:

- Single Responsibility Principle (SRP)- A class should have only one responsibility or purpose. This makes it easier to maintain and modify.
- Dependency Inversion Principle (DIP)-High level modules depend on abstraction instead of concrete classes, this increases flexibility and increases maintainability of the code.

(c) Explain how these principles improve software scalability and maintainability.

ANSWER: Abstraction, Encapsulation and Information handling helps to hide the code from unauthorized users, these properties make the design more modular and reusable which in turn makes the software more scalable and maintainable.

(5 Marks)

3. Suppose a software company is developing a **Smart Pharmacy Management System**.

(a) Explain the phases of the Software Development Life Cycle (SDLC):

- Feasibility Study- This phase checks whether the project is practical and financially possible. It helps decide if development should continue.
- Requirement Analysis- The needs and expectations of users are collected and documented. These requirements guide development.
- System Design- The overall structure and architecture of the system are planned. It acts as a blueprint for development.
- Implementation- Developers write the program code based on the design. All features are developed in this phase.
- Testing- The software is tested to identify and fix errors. It ensures the system works correctly.
- Maintenance- After deployment, the software is updated and bugs are fixed. New features may also be added.

(b) Describe the significance of each SDLC phase.

ANSWER: Each SDLC phase is developed in an organized manner. So, the whole project is reliable and it also reduces the faults that may occur in the development

(c) Explain how SDLC helps reduce project risks and improve software quality.

ANSWER: SDLC reduces project risks through proper planning and testing. It also improves software quality by following a structured process.

(5 Marks)

4. Consider a project for developing a **Smart E-Commerce System**, where customer requirements frequently change due to market demands.

(a) Explain the following SDLC models:

- Waterfall Model- The Waterfall Model follows a sequential process where each phase is completed before moving to the next.
- V-Model- The V-Model performs testing alongside every development phase. It improves

- software quality through early testing.
- Incremental Model- The Incremental Model develops the software in small parts. Each increment adds new functionality.
- Agile Model- The Agile Model develops software in short iterations with continuous customer feedback. It easily handles changing requirements.

(b) Compare the Waterfall Model and Incremental Model.

ANSWER: The Waterfall Model follows a fixed sequence and is difficult to change. The Incremental Model develops software in parts and allows easier modifications.

(c) Recommend the most suitable SDLC model for this project with justification.

ANSWER: The Agile Model is the best choice because customer requirements change frequently. It supports flexibility, continuous feedback, and faster updates.

(5 Marks)

5. Suppose a software analyst is designing a **Smart Hospital Information System**.

(a) Explain the role of UML in object-oriented software design.

ANSWER: UML (Unified Modeling Language) is used to visually represent the design of a software system. It helps developers understand and communicate system structure.

(b) Describe the following UML diagrams:

- Use Case Diagram- It shows the interaction between users and the system. It identifies the system's main functions. The main components include actors and use cases.
- Sequence Diagram- It shows the order of messages exchanged between objects over time. It explains how operations are performed. It shows at a particular time what action was done and then how did the flow of actions take place across time.
- Communication Diagram- It shows how objects communicate with each other. It focuses on relationships and message flow. If there is a message or a request then there should be some kind of response, otherwise the communication is broken.
- Activity Diagram- It represents the workflow of activities in a process. It shows the sequence of actions from start to finish.

(c) Explain how UML diagrams support software analysis, design, and documentation.

ANSWER: UML diagrams help analyze, design, and document software systems. They improve communication among developers and stakeholders.

(5 Marks)

6. Consider developing a **Smart Railway Reservation System**.

(a) Explain the concept of object-oriented software design.

ANSWER: Object-oriented software design organizes software using classes and objects. It makes programs modular, reusable, and easier to maintain.

(b) Compare the structured software development approach with the object-oriented software development approach.

ANSWER:

Structured Software Development

1. Focusses on functions and procedures.
2. It has more complexity.
3. Moving from one phase to another is difficult.

Object-Oriented Software Development

1. Focusses on objects and methods.
2. It is less complex when compared to structured development.
3. Moving from one phase to another is quite easier.

(c) Explain how object-oriented design improves software flexibility, scalability, and reusability.

ANSWER: Object-oriented design improves flexibility by allowing easy modifications. It also supports scalability and code reuse through inheritance and encapsulation.

(5 Marks)

RUBRICS FOR CALCULATION

SHORT ANSWER TEST

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	20%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps